
Replication and Predictable-Drift Diagnostics for FTS-Diffusion

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Abstract

1 We revisit FTS-Diffusion (ICLR 2024), a financial time-series generator built from
2 scale-invariant subsequence clustering (SISC), pattern-conditioned diffusion, and
3 learned pattern evolution, and try to reproduce it from the released code. In our
4 runs the released sampling pipeline is near-degenerate: synthetic S&P 500 paths
5 are one repeated motif tiled into an almost perfectly linear price ramp (linear-fit
6 $R^2 \approx 1.0$), independent runs are nearly identical, and synthetic return volatility is
7 about $20\times$ too small. We trace this to three choices in the released sampling code:
8 a deterministic argmax rollout of the pattern chain, a one-unit decoder hidden state,
9 and a commented-out magnitude normalization. We are explicit about confounds
10 we cannot rule out: no checkpoints are shipped, and our training budget is reduced.
11 Consistent with degeneracy, a continuous-rollout Train-on-Augmentation-Test-on-
12 Real (TATR) setup appears to improve S&P 500 forecasting by up to 85%, yet
13 the same protocol never beats its baseline on GOOG or ZC=F. This non-transfer
14 is what a coincidental scale match would produce, not useful augmentation. On
15 the released artifacts we also do not reproduce the authors' Kolmogorov–Smirnov
16 and Anderson–Darling return tests, nor the Appendix B.3 SISC toy targets. As a
17 predeclared extension we add a predictable-drift diagnostic that calibrates on real
18 assets, tracks exploitable artificial alpha in a 500-seed experiment, and improves
19 a 1000-seed augmentation-selection benchmark. We frame these as observations
20 from one reproduction attempt, not a verdict on the method.

21 1 Introduction

22 Deep generative models are attractive for finance because historical market data are scarce, noisy,
23 nonstationary, and not experimentally reproducible. Financial returns also exhibit heavy tails, weak
24 raw-return autocorrelation, volatility clustering, and recurring shapes at irregular time scales [Cont,
25 2001]. Sequence generators try to preserve temporal structure with adversarial, optimal-transport, and
26 score-based objectives [Yoon et al., 2019, Xu et al., 2020, Tashiro et al., 2021]; financial generators
27 additionally need to respect stylized facts and economically meaningful dependence [Wiese et al.,
28 2020]. FTS-Diffusion [Huang et al., 2024] addresses this by discovering variable-length motifs and
29 generating new trajectories motif by motif.

30 This is a replication and diagnostic-extension study, not a new generator. We test whether the
31 published downstream gains are robust to rollout and evaluation details, whether the SISC toy-data
32 targets are reproducible, and whether a simple predictable-drift statistic catches artificial alpha that
33 marginal stylized facts miss. The architecture is well motivated, but the released sampling pipeline,
34 in our hands, produces degenerate artifacts: the synthetic series collapses to a near-linear price ramp
35 built from one repeated motif (Section 4). Under that lens the apparent S&P 500 gain looks less like
36 robust augmentation than a coincidence that does not transfer across assets. We cannot fully separate

37 this from a known confound: no trained checkpoints are released, and our training budget is shorter
38 than the released defaults.

39 We therefore extend the evaluation with a predictable-drift diagnostic. Under a fixed information
40 vector

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)^\top, \quad (1)$$

41 we measure whether the next return r_{t+1} has a nonzero linear conditional mean under a bounded test
42 class. This is not a proof of market efficiency or absence of arbitrage. It is a finite, reproducible check
43 for an important failure mode: a synthetic generator can match tails and volatility clustering while
44 introducing a simple, exploitable conditional drift.

45 **Contributions.** We make five contributions. (i) A code-level diagnosis of the released sampling
46 pipeline (Section 4): three specific choices that drive the synthetic series to a near-degenerate linear
47 ramp, plus the underspecifications we had to resolve. (ii) A six-seed S&P 500 TATR study, with the
48 cross-asset observation that the continuous-rollout gain does not transfer to GOOG or corn futures
49 (ZC=F), which we read as a scale coincidence. (iii) A return-distribution check that, on the released
50 artifacts, does not reproduce the authors’ Kolmogorov–Smirnov and Anderson–Darling results. (iv)
51 A documented Appendix B.3 SISC toy-data gap. (v) A fixed predictable-drift diagnostic, calibrated
52 on real assets, applied to ordered FTS-Diffusion samples, validated by a 500-seed predictive-alpha
53 experiment, and used in a 1000-seed augmentation-selection benchmark. Throughout, we separate
54 claims about the released artifacts from what would need the authors’ checkpoints to settle.

55 2 Background: FTS-Diffusion

56 FTS-Diffusion decomposes financial time-series generation into three modules [Huang et al., 2024].
57 Scale-invariant subsequence clustering (SISC) segments a univariate series into variable-length motifs
58 using dynamic time warping (DTW) [Sakoe and Chiba, 1978] and DTW barycenter-style centroid
59 updates [Petitjean et al., 2011]. Each segment is summarized by a pattern identity p_m , a duration
60 scale α_m , and a magnitude scale β_m . A pattern-conditioned diffusion model then generates concrete
61 segments from motif conditions using a DDPM-style denoising objective [Ho et al., 2020].

62 The long-horizon behavior is controlled by the pattern-evolution module, which learns

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}), \quad (2)$$

63 with a classification loss for the next pattern and regression losses for the next duration and magnitude
64 scales. This separation is useful, but it also creates a sensitive interface: downstream results depend
65 on how the learned state chain is initialized, reset, and sliced into synthetic training blocks.

66 3 Replication protocol

67 **Assets and splits.** The main datasets are daily Yahoo Finance close-price histories for S&P 500,
68 GOOG, and ZC=F. The S&P 500 setup spans 1980–2020 and yields 10,088 trading days and 2,282
69 post-SISC segments with $K = 14$. GOOG spans 2004–2020 with 3,776 trading days and 733 post-SISC
70 segments with $K = 11$. ZC=F spans 2000–2020 with 4,764 trading days and 629 post-SISC
71 segments with $K = 11$. We follow a chronological train/test design: motif extraction, fitted generator
72 components, standardizers, and downstream predictors are fit only on their allowed training windows,
73 and final scores are computed on held-out real data.

74 **Downstream benchmark.** The main downstream benchmark is TATR: Train on Augmentation,
75 Test on Real. The held-out real slice is divided into an initialization portion and a final real test portion.
76 A one-day-ahead LSTM forecaster is trained on the real initialization portion plus Y synthetic trading
77 years, where one synthetic year has 252 days and Y is swept across a fixed grid. The score is mean
78 absolute percentage error (MAPE) on the final real test portion. We use paper-like LSTM settings:
79 one-day ahead, window size 64, hidden size 32, two layers, MAE training loss, Adam with learning
80 rate 10^{-2} , and 100 epochs for the primary S&P 500 runs.

81 **Synthetic rollout variants.** We compare three ways to produce the synthetic augmentation path.

- 82 • **Independent fixed blocks:** generate independent 252-day blocks from the same fixed initial
83 state.
- 84 • **Continuous chunked:** generate one continuous trajectory with the pattern-evolution chain,
85 then split it into 252-day chunks for downstream training.
- 86 • **Continuous with scaler refit:** generate a continuous trajectory and refit the downstream
87 scaler on the augmented training series.

88 These protocols use the same broad data budget but different Markov-chain connectivity. The
89 distinction matters because the pattern-evolution module is the only component that carries long-
90 horizon state across generated motifs.

91 **Compute.** The primary six-seed S&P 500 downstream batch used stored SISC artifacts and PG-
92 M/PEM checkpoints and recorded 10,478 seconds on an Apple M4 laptop with 10 CPU cores and
93 16 GB memory. The lightweight drift diagnostics used the same machine: the ordered FTS audit
94 recorded 629.82 seconds with a four-thread cap, the 500-seed predictive-alpha script recorded 2.83
95 seconds, and the 1000-seed selection benchmark recorded 32.69 seconds. These times exclude earlier
96 exploratory runs not used for the reported tables.

97 4 Degeneracy of the released generation pipeline

98 We reproduce FTS-Diffusion end to end from the released code. Before downstream scores, we report
99 what the generator actually produces, since this conditions every later interpretation. In our runs the
100 released sampling path is near-degenerate: the synthetic price series is a single repeated motif tiled
101 into an almost perfectly straight line. We give three jointly responsible choices, with file references
102 into our working copy (fts-diffusion-ref/), then show the degeneracy on the saved trajectories
103 and the return distribution. These are observations about the released artifacts under our setup, not
104 proofs about the method; Section 4.5 lists the confounds we cannot rule out.

105 4.1 Three choices in the released sampling code

106 **(1) Deterministic argmax rollout of the pattern chain.** The pattern-evolution module (PEM) is
107 rolled out by taking argmax for the next pattern and the next length, and the raw regression mean for
108 the next magnitude, with no sampling (fts-diffusion-ref/models/sampling.py, lines 30, 33,
109 34):

```
110 pred_pattern = torch.argmax(probabilities_pattern, dim=1).unsqueeze(0)
111 pred_length = torch.argmax(probabilities_length, dim=1).unsqueeze(0) + l_min
112 pred_magnitude = pred[:, n_patterns+range_length:].float()
```

115 An autoregressive map $\text{state}_{m+1} = f(\text{state}_m)$ that is deterministic must, iterated from a fixed initial
116 state, converge to a fixed point or a short cycle. The rollout policy (greedy vs. sampling) is not
117 specified in the paper, so this is a defensible reading of the release, but it removes the stochasticity
118 the Markov-chain description suggests.

119 **(2) One-unit decoder hidden state.** The Scaling-AE decoder is an LSTM whose hidden di-
120 mension is fed from `pgm_params['sae_hidden_dim']`, set to 1 in the released configuration
121 (`model_params.py`; the decoder LSTM is defined in `scaling_autoencoder.py`). With a single
122 hidden unit the decoder has almost no capacity, and we observe it collapsing to a fixed output that
123 ignores both the conditioning pattern and the diffusion latent.

124 **(3) Commented-out magnitude normalization.** The sampling step that would rescale the gener-
125 ated shape to unit range before multiplying by the magnitude β is commented out (`sampling.py`,
126 lines 53–55), so the emitted segment has magnitude $\beta \times$ (whatever range the decoder produced)
127 rather than β . In our runs that range is on the order of 0.07, far below the intended scale, which
128 removes the magnitude information β was meant to carry.

4.2 What the saved trajectories look like

Figure 1 contrasts a real S&P 500 path with one released-pipeline synthetic path. The synthetic series is visually a straight line; a least-squares linear fit gives $R^2 \approx 0.99999$. Three quantitative observations accompany the figure. First, independent runs of a configuration are nearly identical, consistent with a deterministic rollout. Second, the PEM state sequence collapses (Table 1): on S&P 500 a single pattern is used in 99.6% of segments (a fixed point), while GOOG and ZC=F settle into period-2 cycles. Third, inspecting the trained PGM decoder pattern by pattern, the decoder output is near-constant across all patterns (cross-pattern standard deviation < 0.001): an up-trend pattern and a down-trend pattern yield essentially the same segment. The net effect is that one tiny upward-creeping segment is tiled across the horizon, which is exactly the linear ramp in Figure 1.

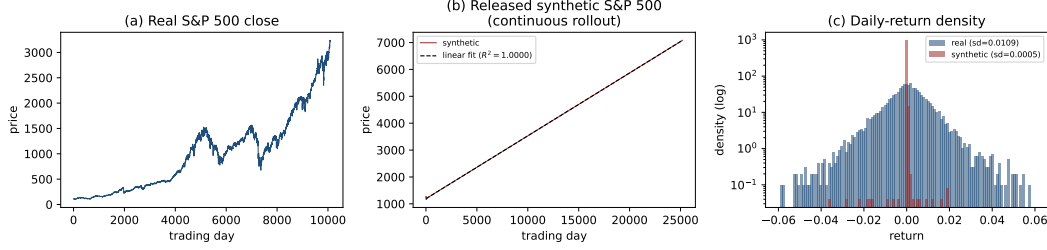


Figure 1: Released-pipeline output on S&P 500. (a) Real close prices show rich structure. (b) One synthetic continuous-rollout path is almost perfectly linear (linear-fit $R^2 \approx 0.99999$). (c) Daily-return density (log scale): synthetic returns concentrate near a single value with standard deviation about $20\times$ smaller than real returns. We show one representative run; runs are nearly identical.

Table 1: PEM state-sequence collapse under the deterministic rollout, from saved `run*_states.npy`. “Dominant pattern share” is the fraction of generated segments using the most frequent pattern.

Asset (K)	Segments	Dominant pattern share	Regime
S&P 500 (14)	2520	99.6%	fixed point
GOOG (11)	2520	50.0%	period-2 cycle
ZC=F (11)	1867	50.0%	period-2 cycle

4.3 The return distribution does not match real

A natural and direct check is the authors’ own return-distribution test. Following their `distribution_tests` routine, we form simple returns and compare real against synthetic with a Kolmogorov–Smirnov (KS) two-sample test and an Anderson–Darling (AD) k -sample test, where a higher p -value means the two return samples are harder to distinguish. The paper reports values consistent with indistinguishability (KS around 0.32, AD around 0.12). On the released artifacts, in our hands, we observe the opposite (Table 2): across 300 windows per asset and protocol the synthetic and real returns are separated in essentially every window, with mean KS p -values at the 10^{-4} to 10^{-7} level and no window passing at $p > 0.05$. The mechanism is visible in the return moments: the released synthetic series has 4 to $60\times$ too little return volatility, because a near-linear ramp has almost no daily variation. We could not reproduce the published goodness-of-fit numbers from the released artifacts.

4.4 Underspecifications we had to resolve

Several choices that matter for the result are not pinned down by the paper. Table 3 lists the ones we encountered, separating the three that are load-bearing for the degeneracy above from the compute-driven confounds and the remaining setup choices. We kept the released values wherever they existed, so that the run reflects the artifact as published rather than our preferences. We note this is an audit aid, not an accusation: underspecification is common, but here it coincides with the failure modes.

Table 2: Return-distribution check on the released artifacts. “Mean KS p ” and “% indist.” are over 300 windows of length 60; higher is more similar (the paper reports KS around 0.32). “Real/Syn. sd” are daily-return standard deviations. `single` is a continuous rollout, `authors` the independent-block protocol.

Asset	Protocol	Real sd	Syn. sd	Mean KS p	% indist.
S&P 500	authors	0.0109	0.0045	9×10^{-4}	0
S&P 500	single	0.0109	0.00046	4×10^{-7}	0
GOOG	authors	0.0184	0.0026	6×10^{-5}	0
GOOG	single	0.0184	0.00029	4×10^{-7}	0
ZC=F	authors	0.0181	0.0028	2×10^{-5}	0
ZC=F	single	0.0181	0.00041	1×10^{-6}	0

Table 3: Paper-vs-release underspecifications we resolved. “Released” marks a value shipped in the released configuration that we kept; “ours” marks a choice we had to make.

Design choice	In paper?	Value used	Note
Scaling-AE decoder hidden dim	no	1 (released)	load-bearing
PEM rollout policy	no	argmax (released)	load-bearing
Magnitude normalization at sampling	implied	commented out (released)	load-bearing
PGM / PEM training epochs	no	30 / 60 (ours; released 400 / 1000)	confound
PCDM diffusion steps	no	30 (ours; released 100)	confound
SISC k -means seed	no	42 (released)	shared seed
GOOG / ZC=F downstream split	S&P only	62.5% train (ours)	cross-asset parity
Initial segment for generation	no	first test segment / random (ours)	protocol switch

157 4.5 Confounds and what we cannot rule out

158 We are explicit that the evidence above concerns the released artifacts as configured and as trained
159 in our hands, not the method in principle. Two confounds prevent a stronger claim. First, no
160 trained checkpoints are shipped with the release, so every reproduction retrains from the released
161 scripts and we cannot compare against the authors’ own generations. Second, our training budget
162 is reduced relative to the released defaults (PGM 30 vs. 400 epochs, PEM 60 vs. 1000, PCDM 30
163 vs. 100 steps), a compute-driven choice that a reviewer can reasonably challenge. The degeneracy is
164 therefore consistent with at least three non-exclusive explanations that we have not fully separated:
165 undertraining, an unlucky single seed, and the one-unit decoder capacity. A full-budget retrain across
166 seeds would falsify or confirm the undertraining hypothesis; we flag it as the natural next experiment
167 and do not assert a method-level conclusion without it. Given how unusual a one-unit decoder hidden
168 state is for a published generative model, it would also be reasonable to ask the authors directly
169 whether that value, and the commented-out magnitude normalization, correspond to the configuration
170 behind the paper’s headline results.

171 5 Replication results

172 5.1 TATR depends on the synthetic-series protocol

173 Our current S&P 500 evidence has three layers. First, the stored result matrices do not support a
174 monotone-downward TATR claim for the documented authors independent-block protocol: the
175 stored S&P 500 authors curve finishes 208.4% worse than its baseline, while the `single` continuous-
176 trajectory curve finishes 71.6% better. Second, a one-seed protocol search found that continuous-
177 rollout variants finish 76.5–85.4% better, whereas independent fixed blocks finish 96.5% worse.
178 Third, Table 4 reports the current primary six-seed long batch.

179 In that long batch, continuous protocols reproduce strong paper-like reductions in downstream MAPE,
180 but independent fixed blocks do not. The conclusion is not driven by a single lucky seed: the final
181 continuous-chunked changes range from -86.8% to -55.1% , while the final independent-block
182 changes range from $+37.1\%$ to $+296.0\%$. The contrast is also visible in Figure 2: continuous

trajectories eventually cross below the real-only baseline, while independent blocks worsen as augmentation increases.

Table 4: Six-seed S&P 500 TATR summary. Percentages are relative to each protocol’s no-augmentation baseline; lower MAPE is better. Ranges are seed-wise final changes.

Protocol	Baseline	Best change	Final change	Final range
Continuous chunked, 1-day	0.0589	−84.9%	−65.2%	[−86.8, −55.1]%
Continuous + scaler refit, 1-day	0.0630	−85.4%	−68.2%	[−84.3, −30.1]%
Independent fixed blocks, 1-day	0.0646	0.0%	+193.2%	[+37.1, +296.0]%
Continuous chunked, 5-day	0.0810	−85.2%	−77.5%	[−85.9, −58.5]%

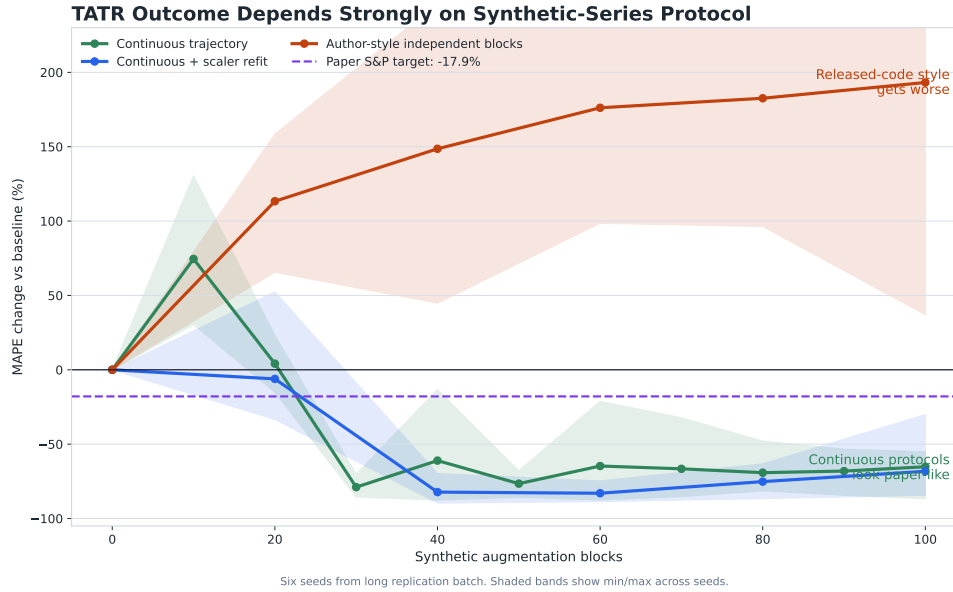


Figure 2: TATR outcome changes sharply with the synthetic-series protocol. Continuous trajectories can look paper-like, while independent blocks worsen with augmentation. The shaded bands show min/max across seeds.

The mechanism is visible in the synthetic price levels. Continuous trajectories start near 1253 and end near 7085 (mean level 4138); independent blocks start from the same level but repeatedly reset, ending near 1249 (mean 1214). The continuous protocol thus drifts its near-linear ramp through the scale of the upward-trending S&P 500 test period, so percentage error can fall simply because synthetic and real price levels overlap, not because the dynamics are informative. Independent blocks stay near the initial level, the overlap never happens, and error grows. Read with Section 4, this is a scale effect, not useful augmentation. The cross-asset evidence below is the cleaner test.

TMTR controls make the replication conclusion more conservative. In the six-seed long batch, the reference TMTR curve reaches a best mean improvement of 26.8% but finishes 359.7% worse; a continuous offset-30 control reaches a best improvement of 50.0% and finishes 17.8% better; a continuous offset-50 control finishes 752.0% worse. Thus the evidence is not “synthetic data always helps.” It is narrower: a particular continuous TATR price-level rollout can create large apparent gains, while other augmentation and evaluation choices do not support the same conclusion.

We make no claim of misconduct; the available generation details do not pin down the downstream result uniquely. We note only that, once the generator is seen to be degenerate, the most economical reading of the S&P 500 gain is a scale coincidence rather than augmentation that works under the right protocol. Either way, a reproducible benchmark should treat the Markov-chain rollout, initialization, block slicing, scaler fitting, and warm-up windows as first-class parameters, and should not rest a positive conclusion on downstream MAPE for a single trending asset.

GOOG and ZC=F test the coincidence reading most directly. We regenerated the stated real-asset SISC and architecture artifacts for both with $K = 11$, $l_{\min} = 10$, and $l_{\max} = 21$. A strict author-faithful downstream TATR rerun is blocked by the released S&P-style split logic, which consumes $252 \times 5 = 1260$ held-out initialization points whereas the GOOG and ZC=F 80/20 held-out windows contain 744 and 963 points, so we adapted the split while keeping the proportions comparable. This changes the absolute MAPE level but not the sign of the augmentation effect, since every percentage is computed within the same split relative to that protocol’s own no-augmentation baseline. Under that within-split comparison the effect is negative on both assets and for both protocol families. The independent-block (authors) protocol finishes 1003.1% worse on GOOG and 75.1% worse on ZC=F. Crucially, the *continuous* rollout that appears to help on S&P 500 does not transfer: its best point over the whole sweep equals the no-augmentation baseline (it never beats it), and it finishes about 106% worse on GOOG and 179% to 431% worse on ZC=F depending on the variant. A protocol that helped because the generator carried useful structure should not flip sign across assets; the continuous gain confined to the one strongly trending asset is the signature of a scale coincidence, not of augmentation value. We therefore treat GOOG and ZC=F as covered for SISC/architecture setup and real-asset diagnostics rather than author-faithful downstream replications, but the within-split sign does not depend on author-faithfulness.

5.2 SISC toy-data replication gap

We also stress-tested SISC on the Appendix B.3 simulated-data setting. The paper target for the multi-pattern toy experiment is average per-unit DTW 0.01 and segmentation Jaccard/IoU 0.784. Across 18 successful runs varying seed, $l_{\max}$, and maximum iterations, our best average per-unit DTW was 0.0321, and our best author-style interval IoU was 0.6418. Figure 3 summarizes the gap.

Appendix B.3 Replication Gap

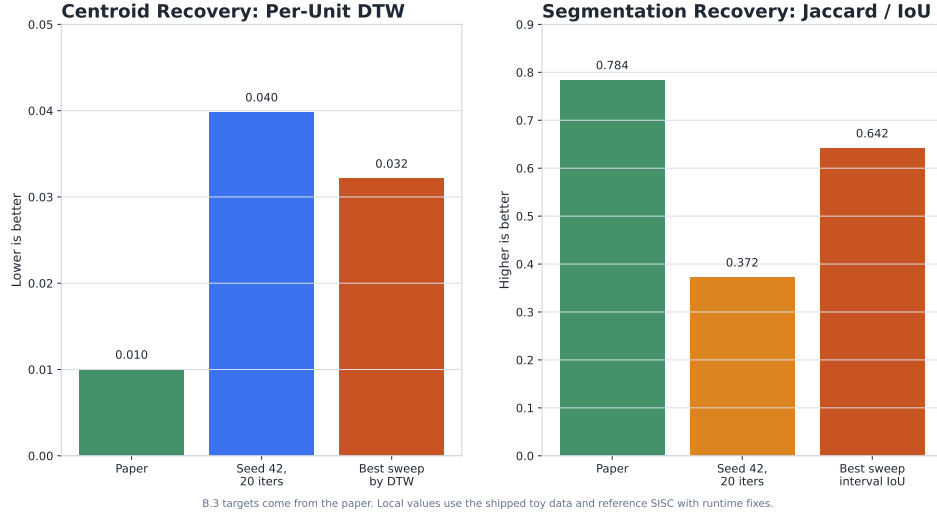


Figure 3: Appendix B.3 SISC replication gap on the toy data. The best replication runs do not reach the reported centroid or segmentation targets.

The gap persisted when increasing SISC iterations from 10 to 20 and when changing $l_{\max}$ from 20 to 21. The best strict boundary Jaccard at tolerance 2 was 0.3608, and the best reference-code-style interval IoU was 0.6418. The best centroid metric remained about $3.2\times$ the reported target. This does not invalidate SISC, but exact B.3 reproduction appears to require more detail about the synthetic generator, seed state, one-pattern construction, and metric definitions. The broader replication lesson is that helper defaults matter: warm-up windows, augmentation scales, plotting conventions, and interval-overlap metrics can all change the apparent result if they are not audited.

6 Predictable-drift extension

6.1 Diagnostic

Stylized facts are necessary but not sufficient for finance. A generator can reproduce heavy tails and volatility clustering while encoding a simple conditional mean, such as $r_{t+1} = \rho r_t + \epsilon_{t+1}$. Marginal tests may not detect this, but a trading signal using r_t can. We therefore treat the drift check as a fixed, predeclared diagnostic rather than a learned discriminator: the chronological split rule, feature map, train-only standardization, null construction, and companion stylized-fact summaries are fixed before evaluating synthetic outcomes.

Motivated by the martingale-difference intuition behind weak-form market efficiency [Fama, 1970], we fix a small information set

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)^\top \quad (3)$$

and a bounded linear class

$$g_b(\mathcal{F}_t) = b^\top \omega_t, \quad \|b\|_2 \leq 1. \quad (4)$$

The population predictable-drift violation is

$$\delta = \sup_{\|b\|_2 \leq 1} |\mathbb{E}[r_{t+1} b^\top \omega_t]|. \quad (5)$$

Let $m = \mathbb{E}[r_{t+1} \omega_t]$. By Cauchy-Schwarz, Equation 5 has the closed form

$$\delta = \|m\|_2. \quad (6)$$

The empirical estimator is therefore

$$\hat{\delta} = \left\| \frac{1}{T-2} \sum_{t=2}^{T-1} r_{t+1} \omega_t \right\|_2. \quad (7)$$

The statistic decomposes into interpretable components: unconditional drift, first-lag return predictability, second-lag return predictability, and two volatility-conditioned mean terms. In our implementation, returns are standardized using training-split mean and standard deviation only; the constant feature remains equal to one. We compare real data, synthetic data, and a block-shuffled null that preserves the return multiset and some local dependence while disrupting longer chronological alignment.

6.2 Real-asset calibration

Before using $\hat{\delta}$ as a rejection statistic, we calibrate it on real held-out assets. For S&P 500, GOOG, and ZC=F, we start from raw close-price CSVs, use the first 80% of simple returns for standardization, and evaluate the last 20%. The null distribution uses 500 replications of 21-day block shuffling on the held-out returns. Table 5 shows that all three real held-out paths lie inside their asset-specific 5–95% null bands, while retaining heavy tails and volatility clustering. This is a useful sanity check: the diagnostic is not simply rejecting real financial data because it is non-Gaussian.

Table 5: Real-asset predictable-drift calibration. The null band is the 5–95% range from 500 block-shuffled held-out paths; the final columns report stylized-fact context.

Asset	Test obs.	Real $\hat{\delta}$	Null 5–95%	ACF($ r $)	Ex. kurt.
S&P 500	2018	0.0297	[0.0222, 0.0303]	0.2165	3.2110
GOOG	755	0.0575	[0.0403, 0.0629]	0.0935	6.1142
ZC=F	953	0.0173	[0.0150, 0.0259]	0.1250	2.0247

6.3 Controlled predictive-alpha test

We next test whether larger $\hat{\delta}$ corresponds to exploitable predictability. In each of 500 seeds, we simulate a zero-mean heavy-tailed GARCH-like path, keep the evaluation-period magnitude path

Table 6: Controlled predictive-alpha test over 500 seeds. Magnitude paths are fixed within seed; only sign dependence changes.

Sample	p	$\hat{\delta}$	OOS corr.	Dir. acc.	Sharpe (sd)
Real evaluation path	—	0.0976	0.0004	0.5025	0.07 (0.77)
IID signs	—	0.0941	0.0036	0.5020	0.09 (0.74)
Markov signs	0.65	0.2088	0.1482	0.5968	2.33 (0.96)
Markov signs	0.75	0.3435	0.2786	0.6952	4.80 (1.05)
Markov signs	0.85	0.5126	0.4112	0.7976	7.72 (1.18)
Markov signs	0.90	0.6164	0.4811	0.8483	9.43 (1.33)

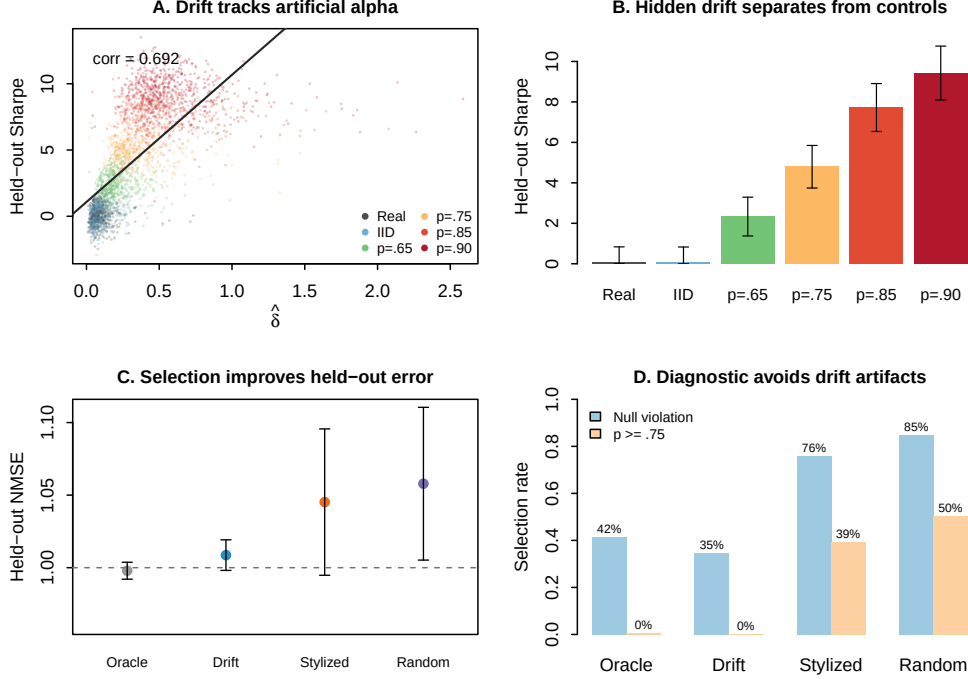


Figure 4: Drift versus no-drift evidence from the larger controlled experiments. The diagnostic tracks held-out artificial alpha, separates hidden-drift sign processes from real and iid controls, and improves augmentation selection by avoiding null violations and high-persistence candidates.

263 $|r_t|$ fixed, and replace only the signs. The iid-sign control has no intentional sign predictability;
 264 Markov-sign samples have sign persistence p . We fit a ridge linear predictor on the first half of each
 265 evaluation path using the same feature map ω_t , then evaluate one-step prediction and a sign strategy
 266 on the second half.

267 Table 6 and Figure 4 show the result. The iid-sign and real controls have near-zero out-of-sample
 268 correlation, near-50% directional accuracy, and near-zero strategy Sharpe. At $p = 0.85$, the hidden-
 269 drift sample has mean $\hat{\delta} = 0.5126$, out-of-sample correlation 0.4112, directional accuracy 79.76%,
 270 and mean annualized Sharpe 7.72 with seed standard deviation 1.18. Across all controlled samples,
 271 pooled $\hat{\delta}$ and out-of-sample Sharpe have correlation 0.6921. Thus the diagnostic is not merely a
 272 reporting statistic; it is aligned with artificial alpha.

273 6.4 Ordered FTS audit and incorporated selection

274 The FTS-Diffusion use case is direct: report Equation 7 on ordered samples from each rollout protocol,
 275 and do not select a protocol only because it improves TATR if its $\hat{\delta}$ lies outside the real/null band.

Table 7: Ordered S&P 500 FTS-Diffusion drift audit over six seeds and 25,200 generated prices per seed. The cutoff is the S&P 500 held-out block-shuffle null 95th percentile, $q_{.95} = 0.0303$.

Protocol	$\hat{\delta}$	$\hat{\delta}/q_{.95}$	Violations	Price std.
Continuous rollout	0.0263	0.87	0/6	1701.5
Independent fixed blocks	0.0370	1.22	6/6	24.5

Table 8: Drift-incorporated augmentation selection over 1000 seeds. Lower normalized MSE is better. High- p is the fraction of selected Markov-sign candidates with persistence $p \geq 0.75$. The oracle uses held-out test MSE and is only a reference upper bound.

Selector	Null viol.	High- p	Styl. score	Test NMSE	Excess
Oracle test-MSE	41.5%	0.5%	0.2135	0.9979	-0.0319
Drift-penalized	34.6%	0.2%	0.0520	1.0087	-0.0211
Stylized-only	76.0%	39.3%	0.0344	1.0452	+0.0154
Random	84.8%	50.2%	0.3106	1.0579	+0.0281

We regenerated six S&P 500 synthetic paths per protocol from stored FTS-Diffusion checkpoints, using 100 ordered 252-day blocks per seed and excluding artificial transitions between independent blocks. Table 7 shows that continuous rollouts stay inside the S&P 500 null band in all six seeds, while independent fixed blocks violate it in all six. The audit therefore does not reject the continuous rollout that produces paper-like TATR gains; it identifies the independent reset protocol as both downstream-harmful and predictably drifted under this statistic.

We next test whether the same principle helps choose augmentations in a controlled benchmark. For each of 1000 seeds, all candidate augmentations shared the same magnitude path but had different sign dynamics. A marginal stylized selector minimized a distance over volatility, tail, quantile, and absolute-return autocorrelation summaries; the drift-incorporated selector minimized

$$d_{\text{styl}} + 0.25 \left[\hat{\delta}/q_{.95}^{\text{real}} - 1 \right]_+^2, \quad (8)$$

where $q_{.95}^{\text{real}}$ is the real block-shuffle null cutoff. The selected augmentation trained the same one-step ridge predictor on a 252-day real initialization plus synthetic returns, and performance was measured on a final held-out real path that the drift-penalized selector does not see.

Table 8 shows that incorporating the diagnostic changes the selected augmentation, not just the reporting. Relative to stylized-only selection, the drift-penalized selector reduces high-persistence choices from 39.3% to 0.2%, cuts real-null violations from 76.0% to 34.6%, and lowers held-out normalized MSE by 0.0365 on average, with a 60.8% paired win rate. If the statistic is incorporated during training rather than selection, use it as a validation-selected penalty

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FTS}} + \lambda \hat{\delta}_{\text{synthetic}}^2. \quad (9)$$

The penalty must be computed on ordered return sequences. Applying it across randomly shuffled mini-batches would destroy the time ordering and should only be logged as a proxy, not as the final diagnostic.

7 Limitations and broader impact

Our replication is constrained by the information and compute available for this study. Most importantly, the degeneracy analysis of Section 4 concerns the released artifacts as configured and as trained in our hands: no checkpoints are shipped, and our training budget is reduced relative to the released defaults. We have not run the full-budget, multi-seed retrain that would separate undertraining from an architectural cause, so we deliberately phrase those findings as observations about the release rather than conclusions about the method. Some runs use reference checkpoints rather than a full fresh retraining of all modules, and the ordered FTS-Diffusion drift audit is limited to the S&P 500 checkpoints and the fixed diagnostic in Equation 7. Train-time validation selection of λ remains to be run before claiming an optimized drift-regularized generator. GOOG and ZC=F have

307 shorter histories than S&P 500, so the downstream split used for S&P 500 is not directly portable
308 without changing the question being asked; we mitigate this by comparing within split, relative to
309 each protocol’s own baseline.

310 The predictable-drift diagnostic is deliberately narrow. It tests a fixed finite-dimensional feature
311 map, not every admissible trading strategy, and it is not a proof of no-arbitrage. Its strength is
312 reproducibility: the statistic is simple, closed-form, and hard to tune after seeing the result.

313 Synthetic financial data can be beneficial for stress testing, augmentation, and reproducibility, but it
314 can also create misleading evidence if artificial predictability is mistaken for real market structure.
315 Any release of synthetic paths should clearly label them as generated data, document the training
316 period and generation protocol, and avoid implying that downstream forecasting gains translate into
317 tradable performance.

318 8 Conclusion

319 FTS-Diffusion has an appealing inductive bias for financial time series: variable-length motifs,
320 pattern-conditioned diffusion, and learned motif evolution. Our reproduction attempt, however, did
321 not yield a working generator from the released artifacts. In our hands the released sampling pipeline
322 collapses to a near-linear ramp built from one repeated motif, the authors’ own return-distribution
323 test does not reproduce on those artifacts, and the one downstream setting that appears to improve
324 forecasting does so only on a single strongly trending asset and fails to transfer to GOOG or ZC=F.
325 The most economical reading of that pattern is a scale coincidence rather than useful augmentation.
326 We are careful not to overstate this: we cannot yet separate the degeneracy from undertraining without
327 the authors’ checkpoints and full training budget, so we present empirical observations about the
328 release, not a verdict on the method. What the evidence does support is a methodological point.
329 Realistic-looking paths and a favorable downstream MAPE on a trending series are weak evidence on
330 their own. Future synthetic-finance benchmarks should report exact rollout protocols, direct generator
331 diagnostics such as the return-distribution and predictable-drift checks used here, and cross-asset
332 transfer, rather than resting positive conclusions on a single asset.

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